Homework 4 – Duality

Due: 13:30, November 12, 2025

1. Find the dual of the linear program.

$$\begin{array}{lrrrl} \min & 14x_1 & -6x_2 & +3x_3 & \\ \text{s.t.} & -3x_1 & +x_2 & +4x_3 & \leq 12 \\ & 6x_1 & +7x_2 & -x_3 & \geq 17 \\ & 2x_1 & & +x_3 & = 10 \\ & x_1 & & & \geq 0 \\ & & x_2 & & \leq 0 \\ & & & x_3 & u.r.s. \end{array}$$

2. Consider the linear programming problem (primal) of minimizing $c^T x$ subject to $Ax = b,\ x \geq 0$. Let x^* be an optimal solution to the primal, assumed to exist, and let π^* be an optimal solution to the dual.

   a) Let $\hat{x}$ be an optimal solution to the primal, where c is replaced by some $\hat{c}$. Show that $(\hat{c} - c)^T(\hat{x} - x^*) \leq 0$.

   b) Let $\tilde{x}$ be an optimal solution to the primal, where b is replaced by some $\tilde{b}$ (the cost vector fixed at c). Show that $(\pi^*)^T(\tilde{b} - b) \leq c^T(\tilde{x} - x^*)$.

3. Consider the linear program (primal problem)

$$\begin{array}{lrrrl} \max & 4x_1 & -x_2 & +3x_3 & \\ \text{s.t.} & 5x_1 & -x_2 & +2x_3 & \leq 4 \\ & -2x_1 & +x_2 & +x_3 & \leq 3 \\ & x_1, & x_2, & x_3 & \geq 0. \end{array}$$

   Using complimentary slackness, check whether $\pi = \left(\frac{4}{3}, \frac{1}{3}\right)$ is an optimal solution to the dual of the linear program. Also, find an optimal solution of the primal problem.

Submission requirements:

1. For each (sub)problem, name the solution file as "*problem_x.ext*," where "*x*" represents the (sub)problem number (*x* = 1, 2, 3 or *x* = 1a, 1b, 1c) and the file extension "*ext*" depends on the file type (Word, Excel, PDF, etc.). If the solution to a (sub)problem contains multiple files (e.g., a Python package), organize the file(s) into a folder and name the folder as "*problem_x*."

2. Pack all the "(sub)problem" folders in a zip file and name the zipped file "*hw_##_中文姓名.zip*," where "##" (two digits) represents the homework number, for example, "*hw_04_赵磊.zip*."